CENT ISLE

Structural Portfolio Simulation for Building Financial Intuition

White Paper

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ABSTRACT

Financial literacy is widely recognized as an essential life skill, yet most financial education still fails to build true intuition. Courses teach definitions and formulas, apps track spending, and simulators reward speculation. None of them place you inside a financial system and let you watch how it works.

Cent Isle is an interactive simulation platform built around a structural metaphor: players construct portfolio structures from asset-class building materials, then observe how those structures hold up under simulated macroeconomic pressure such as inflation, interest rate shifts, and sector downturns. The experience is intentionally short and repeatable. A single Match runs one to two minutes. The goal is not to teach the right answer but to give players enough repetitions that patterns start to click on their own.

As *Cent Isle* enters a global EdTech market valued at $142 billion and a financial literacy education segment growing at 7–9% annually, it also captures something rarely available in financial research: decision-making data from controlled scenarios. Every allocation choice, every adjustment, every response to an economic shock is observable. When aggregated at scale, that behavioral record could prove valuable to institutions that study how ordinary people reason about risk. Importantly, a simulation that teaches inaccurate intuitions is worse than no education at all, which is why calibration is not secondary here; it is foundational to everything *Cent Isle* is trying to do.

The current prototype was validated by senior executives in the financial industry and tailored to meet the needs of diverse markets and users around the world. It establishes the core simulation architecture and the Match framework that everything else runs on. This paper describes the problem it is designed to address, how the simulation works, the market opportunity it targets, and what the platform could become.

The Financial Literacy Problem

Most people handling real financial decisions including retirement accounts, investment portfolios, and debt are doing so without a working mental model of how any of it behaves over time. The OECD's *2020 International Survey of Adult Financial Literacy* found that fewer than half of adults in participating economies could correctly answer basic questions on diversification, inflation, and interest compounding.[1]

Research by Lusardi and Mitchell has consistently documented this gap, finding that financial illiteracy is widespread even in high-income countries and that it correlates directly with poor retirement planning, over-indebtedness, and inadequate risk management.[2]

Part of what makes this hard to fix is that financial systems are dynamic and don't sit still long enough to memorize. They respond to inflation, interest rate changes, sector news, and a dozen other variables at once. Understanding them isn't a knowledge problem, it's closer to a pattern recognition problem. You develop it the same way you develop anything else that requires judgment: by doing it repeatedly, watching what happens, and adjusting accordingly.

Traditional financial education skips this part when it hands you the concepts and leaves you to figure out the intuition on your own, usually in situations where the stakes are real and the feedback loop is years long.

Why Existing Tools Fall Short

There's no shortage of financial education resources. The problem is that most of them address the wrong layer of the learning challenge.

Classroom instruction treats financial concepts as things to be learned and tested. Students can define compound interest and explain diversification. What they usually can't do is feel what a concentrated portfolio looks like when a sector turns, because they've never seen it happen in a context where they were paying attention.

Personal finance apps focus on short-term money management rather than investment behavior. They're genuinely useful for what they do, but they're not teaching anyone to think about portfolios.

Investment simulators are closer to the right idea but tend to pull in the wrong direction. Most reward stock-picking and short-term return maximization. Players learn to focus on what went up, not on what their portfolio structure would look like if three sectors corrected simultaneously. That's a different skill, and it's the one most people actually need.

Spreadsheet models and financial calculators let you explore how variables interact, but they require enough prior mathematical literacy that many learners are locked out before they start. There's a real difference between being able to use a tool and being able to think with it.

None of these are wrong exactly. They're just aimed at the knowledge layer when the real problem lives in the intuition layer.

Experiential Learning as the Right Model

The fields that deal with dangerous complex systems have already figured out how to teach intuition: simulation. Pilots train in flight simulators before they touch a real aircraft. Surgeons practice in controlled environments. Engineers run structural models before committing to a design. All of these environments let you observe consequences without paying the real-world cost of the mistake. Kolb's experiential learning cycle—experience, reflection, conceptualization, experimentation—describes exactly the kind of iterative engagement that simulation enables.[3]

Financial decision-making shares the same properties. Investment outcomes emerge from multiple interacting variables over long timeframes. You can't get good at reasoning about them from static instruction alone. You need to experiment, observe, and repeat. The feedback loop has to be short enough that you can actually learn from it.

Game-based simulations introduce better conditions for learning for a specific reason: they can compress time, amplify feedback, and remove the fear of loss that usually stops people from experimenting with real money. A player who loses a two-minute Match is more likely to try something different than a person who watched their savings drop in a market correction.

Cent Isle: A Structural Simulation of Investing

The central design decision in *Cent Isle* is the structural metaphor. Investing is presented not as a series of trades or numerical targets, but as a physical construction problem. Players build structures from materials that represent different asset classes. Each material has properties that determine how well it handles different kinds of stress. The environment the structure sits in represents macroeconomic conditions.

The practical result is that portfolio construction becomes observable in a way that abstract allocation percentages never quite are. A well-diversified portfolio isn't visualized in a pie chart, it looks like a structure that's still standing when the environment turns hostile. A poorly constructed one shows cracks.

The design deliberately avoids rewarding players for picking winning assets. The challenge is resilience, not performance. You're not trying to maximize returns in a benign environment; you're trying to build something that survives an uncertain one. That shift is important because it pushes players toward the concepts that actually transfer to real financial reasoning—diversification, risk distribution, tolerance for volatility—rather than the ones that look good in a simplified simulation.

Outcomes are communicated visually. The simulation engine runs the numbers, but players see structural behavior: materials showing strain, instability spreading, or collapse. There are no dashboards of financial metrics and no numerical outputs to interpret. The physics of the structure do the work, which means the game is approachable by players of any financial background.

The Simulation in Practice

The screenshots below show two states of the same match with a portfolio of "Conservative" assets. In Figure 1, the player has started with a flat wood-plank platform and the status reads "Structure compromised" indicating a failure state. In Figure 2, the player has paused at Year 2 of 6 with a more substantial build underway: stone (bond) and brick (equity) blocks stacked on an expanded foundation, with 49 seconds remaining.

Figure 1: "Structure compromised" — the player's first attempt collapses under rising sea conditions at Year 0.

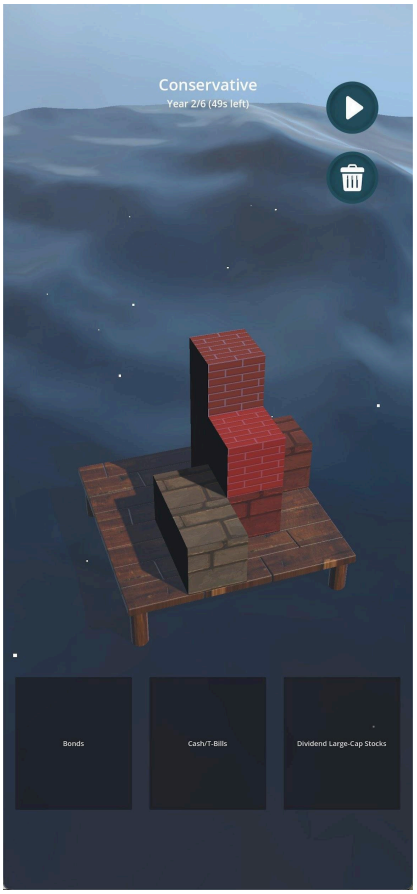

Figure 2: A paused Match at Year 2/6. The player is mid-construction with bonds (stone) and equities (brick) placed on a foundation platform.

The asset cards at the bottom (Bonds, Cash/T-Bills, and Dividend Large-Cap Stocks) represent the Conservative portfolio configuration. Players pull from these cards to build. The rising sea environment is the macroeconomic stress condition for this scenario. The visual clearly communicates structural vulnerability when it exists.

The Match System

Matches are the core unit of play. Each one is a self-contained scenario running one to two minutes. The brevity is intentional, as short sessions lower the barrier to experimentation and make it practical for players to run several Matches in a row, adjusting their approach each time.

At the start of a Match, players choose a portfolio configuration that determines which asset-class materials are available for that round. The environment communicates the economic conditions. Players build, and the simulation runs.

Environmental pressure increases continuously as the Match progresses. Materials accumulate stress. A well-balanced structure distributes that stress across its components and holds. One that's over-concentrated in fragile materials, or missing stabilizing elements, may show gradual strain or sudden cascading failure depending on where the weakness is.

The outcome is binary: the structure survives the scenario, or it doesn't. Players don't need to interpret a score or analyze a metric, they can just watch what happened and decide what they'd do differently next time.

The Simulation Engine

Behind the visual layer, the simulation engine manages the mechanics that determine what actually happens to a structure. Each asset material has a tolerance profile, a capacity to absorb different types of stress. Environmental conditions apply pressure continuously, depleting that tolerance at rates that depend on the match between the asset type and the economic scenario.

When stress is distributed effectively across materials, the structure holds for a long time. When it's concentrated in components that are poorly suited to the current environment, tolerance runs out faster, and failure tends to propagate rather than stay contained.

The engine is intentionally opaque to the player. There are deliberately no numerical thresholds visible, no equations to read, no output beyond what the structure is doing. The goal is intuition, not analysis. A player who can feel that a bond-heavy structure handles inflationary environments poorly, before they could articulate why, has learned something that actually transfers.

Simulation Calibration

A simulation intended to build financial intuition is only useful if it teaches accurate intuitions. A game that rewards portfolios which would actually fail in real economic environments or penalizes those that would hold creates a problem worse than no education at all. The calibration of *Cent Isle*'s simulation is therefore not a secondary concern; it is foundational to the platform's educational credibility.

The current calibration approach is transparent in its method even where it isn't exhaustive in its precision. Each asset class maps to a simplified tolerance profile describing three characteristics: baseline resilience (how well the material holds under neutral conditions), stress sensitivity by condition type (how much each macroeconomic scenario amplifies or dampens stress on that material type), and recovery behavior (whether tolerance is partially restored between stress cycles or depleted linearly).

These profiles are informed by well-documented historical relationships between asset classes and macroeconomic environments and have been reviewed by financial professionals in global markets. Bonds, for example, carry high baseline resilience in low-volatility scenarios but face elevated stress sensitivity under inflationary pressure—a relationship supported by decades of fixed-income performance data. Equity materials reflect higher baseline volatility but greater resilience to mild inflationary environments. Cash and T-Bill equivalents exhibit low stress sensitivity across most conditions but low upside, consistent with their role as stability buffers rather than growth instruments.

The current implementation uses stylized approximations of these relationships rather than live market data or quantitative optimization. The goal at the prototype stage is directionally correct behavior rather than numerically precise simulation. A bond-heavy portfolio should feel stable in most environments but crack under sustained inflation. An equity-heavy one should show more volatility but survive longer in growth environments. Those are the intuitions that transfer.

Future development will expand the calibration to model second-order effects. For example, how a rate shock doesn't just stress fixed-income materials directly, but ripples into equity materials as growth expectations compress. Making these chain reactions legible to a casual player, not just a finance professional, is a core design challenge for the next phase. The platform is also designed to allow scenario-specific calibration, so that named economic environments (inflationary recession, rate-shock correction, sector-specific downturn) can be modeled with profiles informed by the relevant historical episodes.

The calibration framework, including the simulation engine and research methodology, is proprietary. What *Cent Isle* can confirm is that the system is grounded in established asset-class behavior under documented macroeconomic conditions. The educational integrity of the platform depends on this transparency.

Play Mode and Learn Mode

Cent Isle supports two modes of engagement that serve different purposes without requiring different underlying systems.

Learn Mode is structured. Scenarios are designed to isolate specific financial principles; what happens to a bond-heavy portfolio when rates rise, what diversification actually does to stress distribution, how a liquidity buffer changes the outcome in a volatile scenario. Players receive contextual feedback after each Match that connects what they observed to the concept being illustrated. It's meant for audiences who are building foundational understanding.

In Play Mode there are no guided lessons and no explanatory feedback, just economic environments of increasing complexity and the challenge of building something that survives them. The learning that happens here is slower and messier, but it's also more durable. Players who work through enough scenarios develop true pattern recognition without being told what patterns to look for.

The separation also makes *Cent Isle* easier to position in different contexts. Schools and financial literacy programs can build curriculum around Learn Mode. Independent users who want the simulation experience without the instruction can go straight to Play Mode. Neither group needs to use the other mode for the platform to deliver value to them.

Behavioral Data Platform

Every Match in *Cent Isle* generates a record of how a player behaved: what they built, in what order, how they responded when conditions changed, whether they concentrated or distributed risk. When you run across enough players and enough scenario variations, that record begins to look like something financial researchers don't currently have good access to: direct observation of portfolio construction decisions made under simulated economic pressure.

Existing behavioral finance data has real limitations. Survey-based risk profiling captures stated preferences, which diverge from revealed preferences more often than not. Historical market data shows outcomes, but not the reasoning that led to them. There's very little in the literature that directly observes how non-expert individuals construct and adjust portfolios in response to dynamic economic signals. This gap has been noted in the behavioral economics literature as a significant obstacle to understanding retail investor behavior.[4]

Cent Isle's simulation environment is designed to make this observable. Scenarios can be standardized and replicated and economic conditions can be varied systematically. The

decisions players make are logged at a granular level. When aggregated, that dataset has immense value for financial institutions, behavioral researchers, and economic analysts interested in how ordinary people think about risk.

Data Privacy and Ethical Framework

The behavioral data opportunity described above depends entirely on a privacy and consent framework that players, institutions, and regulators can trust. *Cent Isle*'s data practices are built around three principles: transparency about what is collected, meaningful consent before any data is used for research purposes, and strict anonymization before any data leaves the platform.

Consent and transparency

Players are informed at onboarding that aggregate gameplay data may be used for anonymized research into financial decision-making patterns. Participation in data collection is opt-in. Players who decline retain full access to all simulation features. No personal financial data—account information, real investment history, or identifying financial details—is ever collected.

Players who opt in get a clear explanation of what is recorded (structural choices, timing of decisions, response to scenario changes), how long it's retained, and how it's used. That transparency is integral to the platform's educational mission. Understanding that your decisions are generating behavioral insights is itself a form of financial self-awareness.

Anonymization and data handling

All behavioral data is anonymized before any aggregation or external use. Player identifiers are hashed at the point of collection. Individual Match records are not disclosed to any third party. External research access is limited to aggregated, de-identified datasets at a granularity level that can't be reverse-engineered to individual players.

The platform is designed for compliance with applicable privacy frameworks including GDPR for European users and CCPA for California residents. If deployed in educational settings serving minors, COPPA and FERPA requirements will govern data handling for those user segments, with additional parental consent mechanisms in place. Data use in those contexts is limited to platform improvement and does not include external research partnerships.

Ethical research boundaries

Behavioral data partnerships will operate under formal agreements that define permitted research uses, including product development informed by aggregate behavioral insights, while prohibiting individual profiling, retargeting, and uses outside the agreed research scope.

An independent advisory review of the data governance framework is planned before any external partnerships are established. The platform will publish its data use policy publicly and update it when practices change.

Current Development Status

The current prototype represents five months of work by a single developer. The focus during this period has been establishing the core systems that everything else depends on: the Match framework, the simulation engine, the structural tolerance model, and the scenario generation pipeline.

The prototype demonstrates that the structural portfolio metaphor works in practice. Players can construct structures, the simulation applies stress, and the visual feedback communicates what is happening without requiring financial literacy to interpret.

Several supporting systems are in place: portfolio selection, scenario generation, and the asset-condition interaction model that governs how environmental stress affects different materials. Together these form a stable base for the next phase of development.

The current prototype establishes the core architecture with an initial asset library and scenario set. Further development will expand both significantly, adding full asset class coverage, a broader scenario library reflecting distinct macroeconomic episodes, and the visual fidelity needed for institutional deployment. The foundation is in place; the next phase builds the market-ready product on top of it.

Early Prototype Observations

Early prototype testing suggests that the structural metaphor functions as intended for players without prior financial background. The core feedback loop—build, observe failure or success, iterate—is legible without instruction. Players who encountered the "Structure compromised" failure state in early testing consistently showed awareness of what caused it, even without explicit explanation, and adjusted their next build accordingly.

Several design decisions evolved through the prototype period. The initial implementation presented environmental stress as a numerical meter, which players found easier to optimize against than to learn from. Removing that explicit numerical feedback and replacing it with structural behavior alone produced more exploratory play and more varied portfolio construction choices.

Early testing also informed Match duration. Sessions longer than two minutes showed diminishing engagement without improving learning outcomes, while sessions under 60 seconds didn't give players enough time to observe stress propagation. The current one-to-two minute target came out of those observations.

Formal user research with larger sample groups is planned for the next development phase. Metrics under development include session completion rates, strategy variation across repeat plays, and a pre/post assessment measuring shift in conceptual understanding of diversification and risk.

Development and Team

Cent Isle was conceived and built as an independent project under Arques Studio, a solo game development practice focused on building interactive systems at the intersection of education and simulation. The studio's prior work includes projects exploring procedural design, real-time simulation, and game-based learning mechanics.

The founding developer is a student at the University of Minnesota's Carlson School of Management, pursuing a double major in Business Analytics and Management Information Systems with a minor in Medieval Studies, enrolled in the Honors Program. His background spans full-stack software development, real-time simulation engineering, game design, and visual systems, a combination that made him well-suited to build a product that sits at the intersection of financial education, interactive media, and behavioral data infrastructure.

Cent Isle was built entirely independently over five months, without a development team, external funding, or institutional backing. Every architectural decision, every iteration, and every design choice had to be justified against a single question: does this make the simulation work better? The result is a prototype that is lean by necessity and coherent by design. The core systems were not assembled from off-the-shelf components, they were built from the ground up to serve a specific educational theory, and they hold together because the founder understood every layer of what he was building.

Prior to Cent Isle, the founder developed a series of independent game prototypes focused on procedural systems, real-time simulation mechanics, and educational game design, including projects exploring hand-tracking interaction, body-motion engines, and procedural map and lore generation for tabletop applications. These projects established a consistent design philosophy: complex systems become learnable when players can interact with them directly, observe consequences, and iterate without fear of permanent failure. Cent Isle applies that philosophy to financial education.

The next phase of development will involve bringing in collaborators with expertise in financial economics (for simulation calibration validation), UX and visual design (for environment and material fidelity), and educational assessment (for learning outcome measurement). Conversations with potential advisors in these areas are ongoing.

The founding developer is actively seeking institutional partnerships with financial literacy programs, academic researchers in behavioral finance, and EdTech organizations with existing distribution in school and community settings.

Market Opportunity

The financial literacy education market sits at the intersection of three overlapping sectors, each with substantial and growing demand. The global financial literacy and personal finance education market was estimated at approximately $1.4 billion in 2023 and is projected to grow at a compound annual rate of around 7–9% through 2030, driven by rising institutional demand for evidence-based financial education programs.[5]

The broader EdTech market, which *Cent Isle* operates within, was valued at approximately $142 billion globally in 2023 and is projected to exceed $400 billion by 2028.[6] Simulation-based and game-based learning represent a fast-growing segment within this market, with increasing adoption in K–12, higher education, and corporate training contexts.

The behavioral finance research market—the addressable space for *Cent Isle*'s secondary data platform—is harder to size directly, but financial institutions collectively spend tens of billions of dollars annually on consumer insights, behavioral analytics, and investor research. A simulation-derived behavioral dataset capturing portfolio construction under controlled economic conditions occupies a novel position in this landscape.[7]

Institutional demand is also rising from a regulatory perspective. In the United States, the SECURE 2.0 Act (2022) expanded requirements for workplace financial literacy programs,

creating new demand for engaging, evidence-based financial education tools. Similar mandates exist in the UK, Australia, and several EU member states.

Competitive Landscape

Cent Isle competes in the financial education simulation space but differs from existing tools along several dimensions that are hard to replicate without rebuilding around a fundamentally different design philosophy.

Platform	Experiential learning	Macro modeling	Resilience focus	Behavioral data	Literacy barrier
Cent Isle	✓	✓	✓	✓	None required
Investopedia Simulator	~	–	–	–	Moderate
Stock Market Game (SIFMA)	~	–	–	–	Low–Moderate
Budgeting apps (YNAB / Mint)	–	–	–	–	Low
EverFi / traditional curricula	~	–	–	–	Low
Spreadsheet / calculator tools	–	~	–	–	High

✓ *Full capability* ~ *Partial* – *Not present*

The most direct competitive comparison is to investment simulators like the Investopedia Stock Simulator and SIFMA's Stock Market Game. Both provide interactive market environments, but both are organized around return maximization in simplified markets. They measure success by how much a player's portfolio grows, which incentivizes short-term speculation and teaches players to focus on individual asset performance rather than portfolio structure. Neither captures macroeconomic stress dynamics or generates behavioral research data.

Budgeting apps and personal finance tools occupy a different part of the market. They address day-to-day financial management rather than investment intuition, and their interactive element is transactional rather than experimental. EverFi and similar digital financial literacy curricula are closer to traditional instruction delivered through a screen than to simulation.

Cent Isle's differentiation isn't primarily technological. The structural metaphor, the resilience-over-returns design philosophy, and the macroeconomic modeling layer are choices that reflect a different theory of what financial education should accomplish. That theory—that intuition, not knowledge, is the real gap—is what no existing tool has been built around.

Defensibility

Cent Isle's long-term defensibility does not rest on the structural metaphor alone. The metaphor is the entry point, but what accumulates behind it is harder to replicate.

The behavioral dataset is the platform's most durable competitive asset. Every Match generates a granular record of how a player constructs a portfolio, responds to environmental pressure,

and adjusts after failure. Aggregated across tens of thousands of players running standardized scenarios, they build something that does not exist anywhere else in the financial research landscape: a controlled, replicable record of how ordinary people reason about risk in real time. A competitor who builds a similar simulation two years from now starts with no data. *Cent Isle*, if it scales its player base deliberately, holds an asymmetric advantage that compounds with every session played.

Institutional relationships add a second layer of defensibility that is less visible but equally durable. EdTech products embedded in school curricula, university courses, and nonprofit financial literacy programs are difficult to displace. Switching costs are high as curricula must be rebuilt, instructors retrained, and student cohorts re-onboarded. A platform with ten established institutional partners and measurable learning outcome data carries reference credibility that no amount of capital can manufacture quickly.

Simulation calibration credibility represents a third, slower-building advantage. Developing a simulation that teaches accurate financial intuitions requires time, domain expertise, and a transparent methodology that can withstand scrutiny from financial professionals and academic researchers. *Cent Isle* is investing in that process from the prototype stage rather than treating it as a later-phase concern. A competitor can replicate the visual metaphor in months, but building a calibration framework with documented methodology, expert review, and a track record of directionally correct behavior takes years. The gap continues widens with every validation cycle completed and every scenario environment added to the library.

Scenario library depth compounds in a similar way. Each macroeconomic scenario — inflationary recession, rate-shock correction, sector-specific downturn, stagflation — requires research, calibration, and iterative testing before it teaches what it's supposed to teach. The current library is the starting point. Every scenario added extends the platform's educational range and raises the replication cost for any competitor starting from zero.

Finally, brand trust functions as a moat in a domain where credibility is unusually hard to fake. Financial education is a space where parents, administrators, and institutional buyers are acutely sensitive to who built the tool and how. Cent Isle's commitment to calibration transparency, opt-in data consent, and responsible anonymization practices is being built into the platform's architecture from day one, not retrofitted for compliance. That track record, accumulated over time, is not something a well-funded late entrant can shortcut.

Taken together, these five layers—behavioral data accumulation, institutional relationship stickiness, calibration credibility, scenario library depth, and brand trust—build a defensibility profile that only strengthens with time and scale. The window in which a direct competitor could enter and replicate the full stack is real but finite, and it closes as each of these layers matures.

Business Model

Cent Isle's revenue model is built around two primary streams and one longer-term opportunity. At the current prototype stage, the priority is demonstrating educational value at scale before

optimizing for revenue. The model below reflects a realistic progression from early-stage platform to sustainable business.

Stream 1: Institutional licensing

The primary near-term revenue opportunity is licensing *Cent Isle* to educational institutions and financial literacy programs. Schools, universities, and nonprofits running financial literacy curricula face a consistent problem of engagement with the content. *Cent Isle*'s simulation environment provides something most curricula can't: an interactive system where concepts can be experienced rather than explained.

The licensing model would offer annual site licenses on a per-institution or per-seat basis, with pricing tiers differentiated by institution type (K–12, higher education, nonprofit, corporate training). Based on comparable EdTech licensing models, pricing in this category is structured by institution type: K–12 schools typically license at $2,000–$5,000 per year per campus; higher education institutions at $8,000–$15,000 per year; and enterprise or district-wide deployments at $15,000–$25,000 per year.

Learn Mode is the primary product for this channel. Institutions can deploy scenario libraries aligned to their curriculum, and the platform's structured feedback layer means it can be used independently without significant instructor training. That makes it viable for resource-constrained settings such as community colleges, nonprofit financial literacy programs, and high school economics departments, where instructor-led technology adoption is often a barrier.

Stream 2: Enterprise behavioral data partnerships

The secondary revenue opportunity is the behavioral data platform. Financial institutions, insurance companies, and economic research organizations that study retail investor behavior and financial decision-making represent the target buyer segment for this stream.

The model here is not a data marketplace or a raw data sale, rather it's a research partnership structure. Institutions partner with *Cent Isle* to run standardized scenario sets with agreed player populations and receive de-identified, aggregated behavioral datasets in return. Pricing would be structured as research contracts rather than data subscriptions, with fees reflecting the cost of running the relevant scenarios and the analytical value of the resulting dataset. Early-stage contracts in comparable behavioral research contexts range from $50,000 to $250,000 depending on sample size and study design complexity.

This stream becomes fully operational as the platform scales its player base and scenario library. Early institutional pilots may generate preliminary research datasets suitable for exploratory partnerships, with full-scale contracts unlocking as the behavioral dataset grows in size and demographic diversity.

Stream 3: Consumer premium and scenario packs

A freemium consumer model represents the platform's long-term distribution strategy rather than its primary revenue driver. A free tier provides access to a limited set of scenarios in Play Mode, with premium subscriptions unlocking additional scenario packs, expanded asset libraries, and harder economic environments. This model prioritizes user acquisition and behavioral data volume over near-term consumer revenue.

Consumer subscription pricing in comparable EdTech simulation products typically ranges from $4.99 to $14.99 per month. At a meaningful scale, this stream becomes commercially

significant, but its primary strategic function in the early phase is growing the dataset that makes the enterprise data partnership stream valuable.

Path to revenue

The immediate priority is securing one to three institutional pilot partnerships that demonstrate educational efficacy with real users. These pilots serve two purposes: they generate the user evidence needed to strengthen the behavioral data proposition, and they provide the early revenue that funds continued development. A first institutional partnership at $5,000–$15,000 is achievable within six months of completing the next development phase.

Platform Roadmap

Development is organized into three phases. Each phase has defined deliverables and a rough timeframe. Timelines assume solo development; additional collaborators would compress them.

Phase 1: Depth (months 4-6)

The immediate focus is expanding the simulation's financial depth and visual quality to a level appropriate for institutional pilots. Key deliverables include:

- Expert review and calibration of asset-condition interaction parameters against historical financial data
- Expansion of the asset library to cover all major asset classes (international equities, REITs, commodities, alternatives)
- Development of five to eight new scenario environments covering distinct macroeconomic episodes
- Visual fidelity improvements to materials, environments, and structural animations
- Implementation of the Learn Mode feedback layer with aligned curriculum mapping

Phase 2: Validation (months 6-12)

Phase 2 focuses on formal user research and early institutional deployment. Key deliverables include:

- Structured user study with 100–500 participants measuring pre/post conceptual understanding shift
- First institutional pilot deployments in two to three partner organizations
- Publication of a preliminary findings report on simulation-based financial literacy outcomes
- Implementation of the data collection infrastructure required for behavioral research partnerships

Phase 3: Scale (months 12-18)

Phase 3 targets sustainable institutional revenue and initial behavioral data partnerships. Key deliverables include:

- Expansion of institutional licensing to ten or more partner organizations
- Launch of the first enterprise behavioral research partnership

- Consumer platform launch with freemium tier and premium scenario packs
- Establishment of an independent advisory board including financial economists and educational researchers

Applications

Cent Isle is positioned to serve several different audiences, and the dual-mode design makes it possible to serve them without forcing them into the same experience.

In educational contexts—high school economics courses, university personal finance programs, financial literacy nonprofit curricula—Learn Mode provides a structured complement to traditional instruction, as the concepts introduced in a classroom can be experienced in simulation. The combination of conceptual definition and hands-on experimentation tends to produce better retention than either alone.

Financial literacy programs targeted at adult learners in community settings have particularly poor access to engaging educational tools. Simulation environments that don't require prior financial or mathematical literacy to operate could fill a gap in how these programs run.

Independent learners who want to develop investment intuition on their own terms are a natural fit for Play Mode. The experience is self-directed, the sessions are short, and the difficulty scales naturally as players develop more sophisticated approaches.

Research organizations and financial institutions interested in behavioral data represent a longer-term application. The dataset *Cent Isle* generates is not a replacement for market data or survey data, but a third type of evidence that captures the decision-making process itself.

Conclusion

Financial decision-making is hard to teach well because the thing people actually need to learn—how financial systems behave under uncertainty over time—can't be conveyed through definitions or formulas. You must personally interact with it.

Cent Isle is built around that observation. The structural metaphor gives portfolio construction a physical form that can be observed and manipulated. The Match system makes the feedback loop short enough to learn from. The simulation engine translates macroeconomic conditions into structural behavior without requiring players to interpret financial metrics. The result is an environment where financial intuition can develop through experience rather than instruction.

The current prototype proves the concept works. The architecture is in place, the core loop runs, and the calibration approach is financially grounded. Early observations from prototype testing suggest the experiential design accomplishes what the theory predicts: players adjust their behavior based on what they observe without needing the behavior explained to them.

The next challenge is building the depth and partnerships that would let the platform work at scale in schools, in financial literacy programs, and eventually as a source of behavioral research data. *Cent Isle* needs financial expertise to improve the simulation calibration, design expertise to close the gap in visual quality, and institutional partners who can connect the platform to the audiences it's designed to serve. This document is the beginning of those conversations.

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